Roll No....................

Mid Semester Examination 2025

Name of the Course: **BTech** *Semester*: ***II***

Name of the Paper: **Engineering Physics** *Paper Code*: **TPH-201**

Time: 90 Minutes Maximum Marks: **50**

Note:

(i) All questions are compulsory
(ii) Answer any one sub questions among **a & b** in each main question
(iii) Total marks in each main question are ten
(iv) Each question carries 10 marks

Q1	(10 marks)	CO1
a)	Calculate the separation between two consecutive bright or dark fringes in the Young's double slit experiment.	
OR		
b)	Determine the resultant path difference in the interference through thin wedge-shaped film.	
Q2	(10 marks)	CO1
a)	Calculate the diameter of third and fourth dark Newton's ring for the incident wavelength 5000 Å and the radius of curvature of convex lens is 100 cm.	
OR		
b)	Discuss any method to determine the distance between two virtual sources in the Fresnel's bi-prism experiment.	
Q3	(10 marks)	CO1
a)	Find the resultant amplitude of the sum of n-harmonic waves using geometrical method.	
OR		
b)	Light of wavelength 4500 Å falls normally on a slit of width 20×10^{-5} cm. determine the angular position of the first two minima on either side of the central maximum.	
Q4	(10 marks)	CO2/CO1
a)	Discuss the single mode and multi mode optical fibers and write their applications.	
	OR	
b)	Calculate the minimum numbers of line in grating which will resolve the lines of wavelengths 5880 and 5883 Å in the third order.	
Q5	(10 marks)	CO2
a)	Find the expression of total internal reflection and discuss the type of fibre optics.	
	OR	
b)	The refractive indices for core and cladding are 1.55 and 1.40 respectively. Calculate (i) critical angle (ii) numerical aperture and (iii) maximum incidence angle.	